

Javier Chiang

Senior Slot Game Mathematician | RTP & Certification Specialist

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Portfolio: <https://slot-game-database.vercel.app>

PROFILE

Senior Slot Game Mathematician with 9+ years of experience designing mathematical models for commercial gaming products, through both full-time roles and cross-border freelance work.

Completed 65+ game math models for company projects and supported game development throughout the entire lifecycle, including RTP design, volatility balancing, simulation validation, and certification preparation. Projects have successfully passed GLI and BMM certification.

Published 42 educational articles and completed 120+ independent research projects (including model development and game analysis) focused on slot mechanics, probability structures, game balancing, and mathematical optimization.

CAREER HIGHLIGHTS

- 9+ Years of Specialized Experience in Slot Game Mathematics
- 65+ Commercial Game Math Models Completed, plus Cross-Border Freelance Experience
- 42+ Educational Articles Published
- 120+ Independent Research Projects (Model Development and Game Analysis)
- Projects Successfully Certified by GLI & BMM
- Multiple RTP Configurations Designed and Validated
- Public Database of 150+ Slot Game Models

CORE EXPERTISE

Game Mathematics: Slot Mathematics, RTP Design, Volatility Modeling, Hit Frequency Analysis, Feature Mathematics, Probability Theory

Analysis & Verification: Monte Carlo Simulation, Statistical Analysis, Mathematical Verification, Large-Scale Simulation Testing

Certification & Compliance: GLI Certification Support, BMM Certification Support, Certification Documentation

TECHNICAL SKILLS

Python, C++, C#, JavaScript, Matlab, Excel

Mathematical Modeling, Statistical Analysis, Simulation Development, Certification
Documentation

PROFESSIONAL EXPERIENCE

Probability Engineer

Gaming Industry (company undisclosed) | Oct 2026 – Present

Overseas Freelance Game Mathematician

Remote | 5+ years of experience

- Worked across countries, teams and game frameworks, delivering custom slot and casino game math models, from RTP and volatility design to simulation results and specification documents.

Senior Game Math Designer

Audere Gaming Co., Ltd., Taichung | May 2023 – Sep 2026

- Designed original slot math models and completed 34 game projects in three years.
- Built an auto-tuning system and a spreadsheet generator that removed most repetitive work.
- Built a math model generator supporting Line, Way and Megaways mechanics.

Game Math Engineer

CC TECH, Taichung | May 2018 – Apr 2023

- Handled market research, mechanic analysis, math model design, spreadsheet modeling and validation programs.
- Completed 31 game projects (27 released) and analyzed 100+ games; one title ranked in the top 15 of the company's 200+ products.
- Sped up simulation 5x+ with parallel computing and built a Unity simulation tool to improve tuning efficiency and cross-team communication.

Game Math Engineer

De Gather Technology Co., Ltd., Taichung | May 2017 – May 2018

- First dedicated game math engineer; built lottery, slot and table-game models and the development workflow from scratch.
- Planned prize ratios and validated simulation results.

Test & Validation Engineer

Yield Microelectronics Corporation, Hsinchu | Oct 2013 – Mar 2017

- Wrote automated test programs, evaluated product characteristics and analyzed anomalies.

- Systematic validation with large test-script and parameter sets, the foundation of my simulation and verification approach.

PUBLICATIONS & RESEARCH

42+ educational articles covering RTP Design, Volatility Systems, Slot Mechanics, Probability Theory, Mathematical Modeling, and Simulation Methodologies.

120+ independent research projects focused on slot mechanics, RTP optimization, volatility balancing, and mathematical innovation.

Blog: <https://slotmathmodel.blogspot.com/>

PORTFOLIO

Slot Game Database

<https://slot-game-database.vercel.app>

Playable Demonstrations

<https://alice90426.itch.io/>

GitHub

<https://github.com/alice90426>

EDUCATION

Master of Statistics

Bachelor of Applied Mathematics

LANGUAGES

Chinese — Native

English — Professional Working Proficiency (Written Communication)